

DAVID MUKURUVA

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EDUCATION

Dartmouth College

B.A., Double Major in Mathematics and Biomedical Engineering

Hanover, NH

Expected June 2027

Thayer School of Engineering

Bachelor & Master of Engineering in Biomedical Engineering

Hanover, NH

June 2028

Coursework: Real Analysis (Honors), Complex Analysis (graduate), Linear Algebra (Honors, proof-based), Probability (Honors), Combinatorial Game Theory, Differential Equations

RESEARCH EXPERIENCE

FitzLab, Thayer School of Engineering – Prof. M. Fitzpatrick

2026 – Present

Undergraduate Researcher – quantum control benchmarking, superconducting microwave platform Hanover, NH

- **Showed reinforcement-learned control outcomes form a mixture distribution, not a Gaussian** — ran a 40-run seed sweep on Dartmouth's SLURM cluster and found 2 of 8 seeds on the hardest target frozen at the do-nothing baseline (verified to 10 significant figures against an independent calculation), a failure mode mean-and-error-bar reporting hides
- **Overtuned a “tight, unimodal” result by removing post-selection:** unconditioned on the ancilla, its dispersion doubled and the per-seed ranking reordered — the conditioning, not the control method, had produced the effect
- Reproduced an internal benchmark's structure to machine precision but not its headline (gap traced to unstated parameters; failure-to-converge to PPO gradient-estimate variance); built the open-source SLURM seed-sweep toolkit the group runs on (public repo)

Blaschke Ellipses – Mathematics, thesis-track (Polymath Jr)

June 2025 – Present

Researcher – Advisor: Prof. D. Williams; PI: Prof. Y. Zeytuncu

Hanover, NH / Remote

- **Confirmed both target problems were still open before committing a research year to them** — audited the complete citation graph of the field's key survey (arXiv:2206.07466) through 2024–2025 results
- Attacking whether the Blaschke curve of $B_3 \circ B_2$ is always an ellipse (degree-6 decomposition, Kippenhahn/Darboux-curve criterion), and constructing explicit odd-order ($n = 5$) Poncelet ellipses by extending a cutoff CMV-matrix / orthogonal-polynomial method established only for even orders
- Co-authored *Finding Ellipses: Blaschke Products, Poncelet's Theorem, and the Numerical Range*, presented with the Polymath Jr group at the Joint Mathematics Meetings 2026; write the supporting numerics in Python/Sage and formalize selected results in Lean

Goods Lab, Thayer School of Engineering

Jan 2024 – Sep 2024

Research Assistant (Computational)

Hanover, NH

- Analyzed single-cell RNA data from gynecological-cancer patients in R (Seurat, CellChat, Bioconductor), building reproducible pipelines and cross-referencing 16 cell identities with Enrichr and patient metadata to surface candidate biomarkers

PROJECTS

QuBlitz | Python, JavaScript, simulation, testing

2025 – Present

- Designed an interactive quantum-mechanics game (built within the Fitzpatrick Lab) for phase interference and Bell-pair entanglement; implemented and unit-tested the physics engine with a passing automated test suite

Aleph-Nought | Next.js (TypeScript), FastAPI, PostgreSQL

2025 – Present

- Built a full-stack STEM learning platform covering six Dartmouth math/physics course syllabi (Vercel/Render) with a symbolic exercise engine and an automated content pipeline (Manim, FFmpeg, Whisper, Plotly; Redis, Cloudflare R2)

TEACHING

Dartmouth Physics Department & Dartmouth Emerging Engineers

Apr 2025 – Present

Teaching Assistant & Lead Physics Tutor

Hanover, NH

- Designed the DEE physics support model now used across the introductory sequence; facilitate 10 weekly labs for 40+ students and mentor peer tutors

TECHNICAL SKILLS

Programming: Python (NumPy/SciPy), Sage, Lean, MATLAB, R, C; SLURM/HPC, Git

Methods: proof-based mathematics, numerical methods, probability, reproducibility & experiment design; SolidWorks, Arduino